

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: plasmids, 4.5 kB, supercoiled.
Species Used	Mouse, J558-L, myeloma		

Before the Pulse

Cell Growth Medium	DMEM + 10% Fetal Calf Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Split on the previous day
		Pre-pulse Incubation	10 min on ice

Wash Solution None

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	0 °C		
Electroporation Medium*	DMEM + 10% Fetal Calf Serum	Cuvette Gap	0.4 cm
Cell Density	10 (7) cells in 300 µl	Voltage	0.25 kV
Volume of Cells	300 µl	Field Strength	0.625 kV/cm
DNA Concentration	4 µg DNA per 10 (7) cells		
DNA Resuspension Buffer	Not given	Capacitor	960 µF
Volume of DNA	4 µl	Resistor	(Pulse Controller) Ω none
		Time Constant	50 msec

After the Pulse
Outgrowth Medium DMEM + 10% Fetal Calf Serum

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

PNAS, **87**: 5788 (1990).

Electroporation worked best for J558L cells - DEAE dextran did not; neither did lipofection.

Outgrowth Temperature	37 °C
Length of Incubation	48 hours
Selection Method or Assay Used	CAT assay
Electroporation Efficiency	Not given
Per Cent Survival	60 %

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